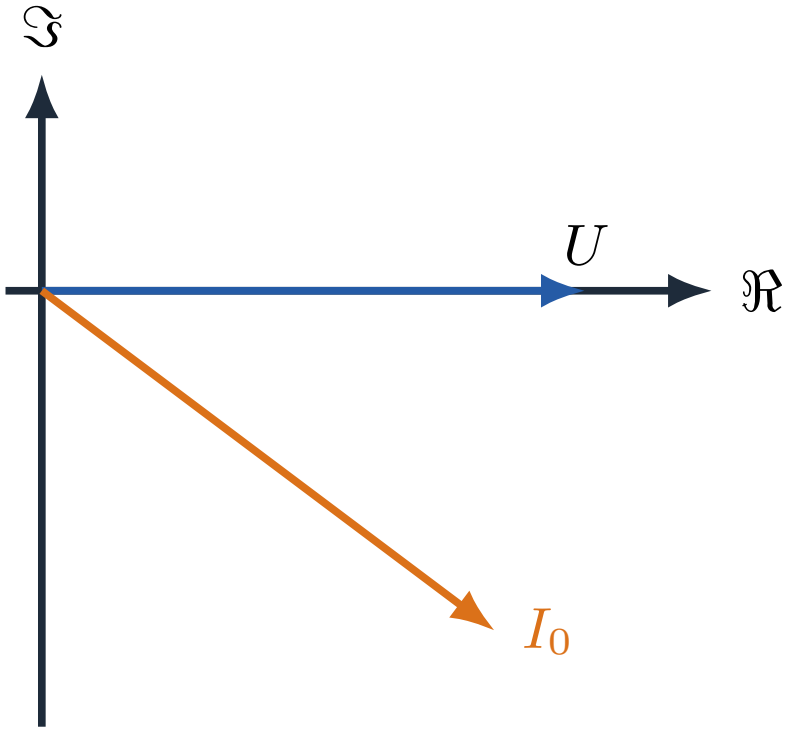
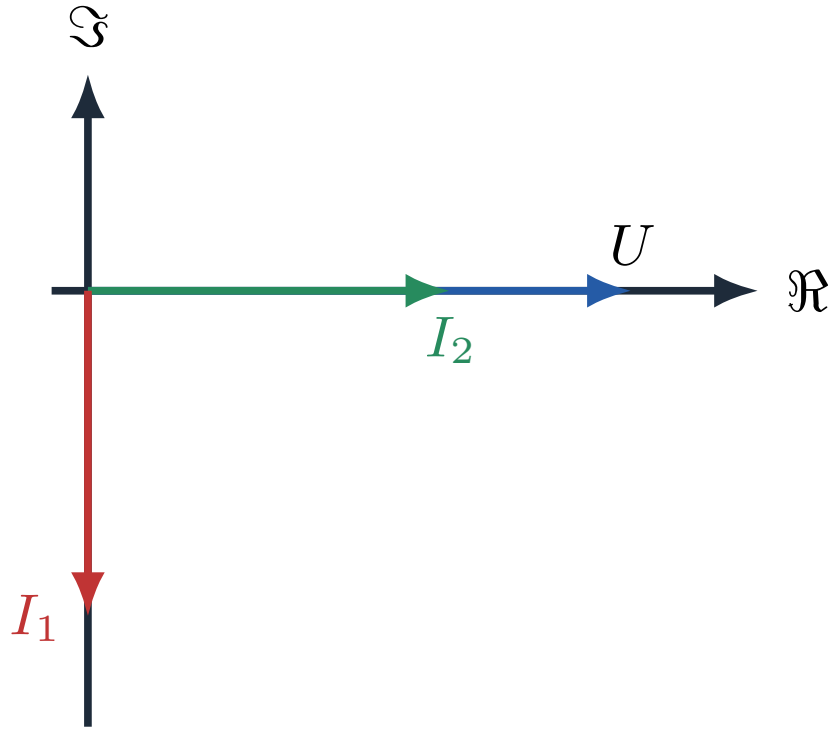


(a) $Z_a = R + 3ir/4$



(b) $Z_b = R + r^2/R$